OBJECTIVE – 1

TO IDENTIFY HOW INVESTORS CHANGE THEIR INVESTMENT BEHAVIOUR DURING PERIODS OF ECONOMIC UNCERTAINTY.

Overall logic : If investor behaviour truly changes during economic uncertainty, then **FII, DII, and Market Turnover should be statistically different during high-uncertainty periods compared to low-uncertainty periods**.

Since one **cannot directly see investors' emotions**. Thus, We **observe their behaviour through market variables**.

STEP 1: IDENTIFY PERIODS OF ECONOMIC UNCERTAINTY

Variable Used

- **India VIX**

India VIX is used as a **proxy for economic and market uncertainty**, as it reflects expected market volatility and investor fear.

STEP 2: DESCRIPTIVE STATISTICS OF INVESTOR BEHAVIOUR

to understand:

- How investors behave **on average**
- How **stable or unstable** their behaviour is
- Whether behaviour appears **volatile during uncertainty**

This step gives a **basic behavioural overview** before comparison and testing.

Variables Used

(Only those relevant to Objective-1)

- **Net FII Investments (₹ Crores)**
- **Net DII Investments (₹ Crores)**
- **Market Turnover**
- **India VIX** *(for context)*

Concepts Used

- **Mean (Average)** → Typical investor behaviour
- **Standard Deviation** → Variability / instability in behaviour

Output:

Metrics	Average	Standard Deviation
VIX	18	6.743266362
FII	-7841.4665	26604.56492
DII	14235.0967	22063.78343
Market Turnover	60297524	67756202.05

Interpretation:

Variable-wise Interpretation

<u>**India VIX (Economic Uncertainty)**</u>

- **Average VIX = 18** indicates that the Indian market experienced **moderate to high uncertainty** during the study period.
- **Standard deviation of 6.74** shows **frequent fluctuations in uncertainty levels**, suggesting that investors were repeatedly exposed to changing risk conditions.

This confirms that **uncertainty is a relevant and varying factor** in your dataset.

<u>**Net FII Investments (Foreign Investors)**</u>

- The **negative average FII value (–7,841.47)** indicates that, on average, **foreign investors were net sellers** during the study period.
- The **very high standard deviation (26,604.56)** reflects **extreme variability** in FII behaviour.

This suggests that **foreign investors react strongly to changing market conditions**, often withdrawing capital during uncertain periods.

<u>**Net DII Investments (Domestic Investors)**</u>

- The **positive average DII value (14,235.10)** shows that **domestic institutional investors were net buyers** on average.
- Although the standard deviation is high (22,063.78), it is **lower than that of FIIs**, indicating relatively **more stable behaviour**.

This implies that **DIIs play a stabilising role** during uncertain market conditions.

<u>**Market Turnover (Trading Activity)**</u>

- The **very high average turnover** indicates **active market participation**.
- The **standard deviation exceeding the average** suggests **sharp spikes in trading activity**, often associated with **panic selling or speculative trading**.

This reflects **heightened trading intensity during uncertain periods**.

STEP-2 Conclusion

The descriptive statistics indicate that the Indian stock market experienced fluctuating levels of economic uncertainty during the study period. Foreign institutional investors exhibited highly unstable and predominantly negative investment behaviour, suggesting risk aversion. In contrast, domestic institutional investors maintained positive and relatively more stable investment patterns. Significant variability in market turnover reflects changing investor risk appetite and heightened trading activity during uncertain market conditions.

STEP 3: COMPARATIVE ANALYSIS

We compare **investor behaviour during High Uncertainty periods and Low Uncertainty periods**.

This step **directly shows behavioural change during uncertainty**.

Basis of Comparison

- **High Uncertainty** → India VIX above average
- **Low Uncertainty** → India VIX at or below average

Variables Compared

- **Net FII Investments**
- **Net DII Investments**
- **Market Turnover**

Output:

Metrics	Normal	Uncertainity
FII	-7814.2747	-7887.955806
DII	16706.9496	10009.02548
Market Turnover	36168901.3	101549684.3

Variable-wise Interpretation:

Net FII Investments (Foreign Investors)

- FII investments are **negative in both periods**, indicating consistent net selling by foreign investors.
- The **slightly more negative value during uncertainty** (–7,887.96) indicates **increased withdrawal during uncertain periods**.

This reflects **heightened risk aversion among foreign investors when uncertainty rises**.

Net DII Investments (Domestic Investors)

- During normal periods, DIIs invest **significantly higher amounts** (16,706.95).
- During uncertainty, DII investments **decline sharply** to 10,009.03.

This suggests that **domestic investors become more cautious during uncertainty**, though they remain net buyers and help stabilise the market.

Market Turnover (Trading Activity)

- Market turnover increases **almost three times** during uncertainty periods.
- This sharp rise indicates **panic-driven trading, portfolio reshuffling, and speculative activity**.

Higher turnover during uncertainty reflects **intensified market activity driven by fear and uncertainty**.

STEP 3 Conclusion:

The comparative analysis reveals a clear change in investor behaviour between normal and uncertainty periods. Foreign institutional investors exhibit increased risk-averse behaviour by marginally increasing net selling during uncertain periods. Domestic institutional investors reduce their investment activity, indicating cautious behaviour, while remaining net buyers. A substantial increase in market turnover during uncertainty periods reflects panic-driven and speculative trading activity in response to elevated market uncertainty.

STEP 4: t-TEST (STATISTICAL CONFIRMATION OF BEHAVIOURAL CHANGE)

We apply a **t-test** to check whether the differences observed in **STEP 3** are **statistically significant**.

STEP 3 showed that investor behaviour **changes during uncertainty**, but that was based on averages.
The **t-test confirms whether this change is real and statistically valid**, not due to random fluctuations.

Variables Tested

We test **separately** for:

- **Net FII Investments**
- **Net DII Investments**
- **Market Turnover**

Each variable is tested between:

- **Normal period**
- **Uncertainty period**

Type of t-Test Used (VERY IMPORTANT)

t-Test: Two-Sample Assuming Unequal Variances

Why this test is appropriate:

- Normal and uncertainty periods are **independent**
- Variances are **unequal**
- Data is **time-based**, not paired

Output:

t-Test

Metric	P(T<=t) two-tail

FII	0.991096308
DII	0.213731297
Market Turnover	0.000236821

Decision Rule (Reminder)

- **p-value < 0.05 → Reject H_0 (Significant change)**
- **p-value ≥ 0.05 → Fail to reject H_0 (No significant change)**

Variable-wise Interpretation

<u>Net FII Investments</u>

- **p-value = 0.9911 (> 0.05)**
- We **fail to reject the null hypothesis**

Interpretation:

The t-test indicates that the difference in FII investments between normal and uncertainty periods is not statistically significant. Although foreign investors show net selling behaviour during uncertainty, the change in average investment levels is not statistically strong.

FIIs are **consistently risk-averse**, but the *magnitude* of change is not statistically different.

<u>Net DII Investments</u>

- **p-value = 0.2137 (> 0.05)**
- We **fail to reject the null hypothesis**

The t-test results show no statistically significant difference in DII investments between normal and uncertainty periods, indicating that domestic institutional investors maintain relatively stable investment behaviour despite market uncertainty.

DIIs act as **market stabilisers**, not panic-driven investors.

<u>Market Turnover</u>

- **p-value = 0.00024 (< 0.05)**
- We **reject the null hypothesis**

The t-test confirms a statistically significant increase in market turnover during periods of economic uncertainty, indicating heightened trading activity driven by panic, fear, and speculative behaviour.
Behavioural change is strongest in trading activity, not net investment values.

<u>STEP 4 CONCLUSION</u>

The t-test results reveal that while changes in FII and DII investment values across normal and uncertainty periods are not statistically significant, market turnover increases significantly during periods of high economic uncertainty. This indicates that uncertainty primarily affects investor trading behaviour rather than the net level of institutional investments.

STEP 5: EVENT-BASED INTERPRETATION

We **link changes in investor behaviour** observed in Steps 2–4 with **major economic and market uncertainty events** mentioned in the dataset.

Month	BSE	NSE	India VIX	FII	DII	Market Turnover	Uncertain Periods
Sep-2018	11418.76	11297.06	15.03486	-9620.56	8663.88	23209438	0
Mar-2019	12090.72	11273.48	15.98961	32371.43	-13930.3	22581720	0
Mar-2020	9000.99	9426.312	53.06357	-65816.7	55595.18	23968172	1
Mar-2021	15372.5	14835.1	20.65	1245.22	5204.42	97975987	1
Jun-2022	16554.1	15933.16	21.12841	-58112.4	46599.23	2.19E+08	1
Mar-2023	18256.53	17226	14.35464	1997.7	30548.77	2525378	0
Mar-2024	23440.8	22187.31	13.68833	3314.47	56311.6	3455293	0

Events:

Month	Event
18-Sep	NBFC Crisis
19-Mar	Volatile/Uncertain
20-Mar	COVID Crash
21-Mar	Post-COVID Recovery
22-Jun	Inflation Shock
23-Mar	Recovery, TowardsStable
24-Mar	Stable Market

Event-wise Interpretation (Based on Your Data)

NBFC Crisis – Sep 2018 (Normal Period)

- **India VIX:** 15.03 (Low)
- **FII:** –9,620.56 (Mild selling)
- **DII:** 8,663.88 (Buying support)
- **Turnover:** Moderate

Interpretation:

During the NBFC crisis, market uncertainty remained relatively controlled. Foreign investors showed limited withdrawal, while domestic investors provided support, indicating stable investor behaviour under moderate uncertainty.

Volatile / Uncertain Phase – Mar 2019 (Normal Period)

- **India VIX:** 15.99
- **FII:** +32,371.43
- **DII:** –13,930.25

Interpretation:

Despite volatility, uncertainty levels were not extreme. Foreign investors displayed opportunistic buying, while domestic investors reduced exposure, reflecting selective risk-taking behaviour.

<u>**COVID-19 Crash – Mar 2020 (High Uncertainty)**</u>

- **India VIX: 53.06 (Extremely High)**
- **FII:** –65,816.70 (Massive outflow)
- **DII:** +55,595.18 (Strong buying)
- **Turnover:** Elevated

Interpretation:

The COVID-19 crash represents the peak of economic uncertainty in the study period. A sharp spike in India VIX coincided with massive FII withdrawals, indicating extreme risk aversion. In contrast, domestic institutional investors absorbed selling pressure, highlighting their stabilising role during crisis periods.

<u>**Post-COVID Recovery – Mar 2021 (High Uncertainty)**</u>

- **India VIX:** 20.65
- **FII:** +1,245.22
- **DII:** +5,204.42
- **Turnover: Very High (97.9 million)**

Interpretation:

Although uncertainty remained elevated, gradual recovery signs led to cautious participation by both foreign and domestic investors. High turnover reflects increased trading activity during market adjustment and confidence rebuilding.

<u>**Inflation Shock – Jun 2022 (High Uncertainty)**</u>

- **India VIX:** 21.13
- **FII:** –58,112.37
- **DII:** +46,599.23
- **Turnover: Extremely High (219 million)**

Interpretation:

The inflation shock triggered renewed uncertainty, leading to sharp FII withdrawals and aggressive DII buying. The surge in market turnover confirms panic-driven and speculative trading behaviour during uncertainty.

<u>**Recovery Phase – Mar 2023 (Normal Period)**</u>

- **India VIX:** 14.35
- **FII:** +1,997.70
- **DII:** +30,548.77

- **Turnover:** Low

Interpretation:

As uncertainty declined, both foreign and domestic investors returned to the market, reflecting improved confidence and stabilised investment behaviour.

<u>Stable Market – Mar 2024 (Normal Period)</u>

- **India VIX:** 13.69
- **FII:** +3,314.47
- **DII:** +56,311.60
- **Turnover:** Low

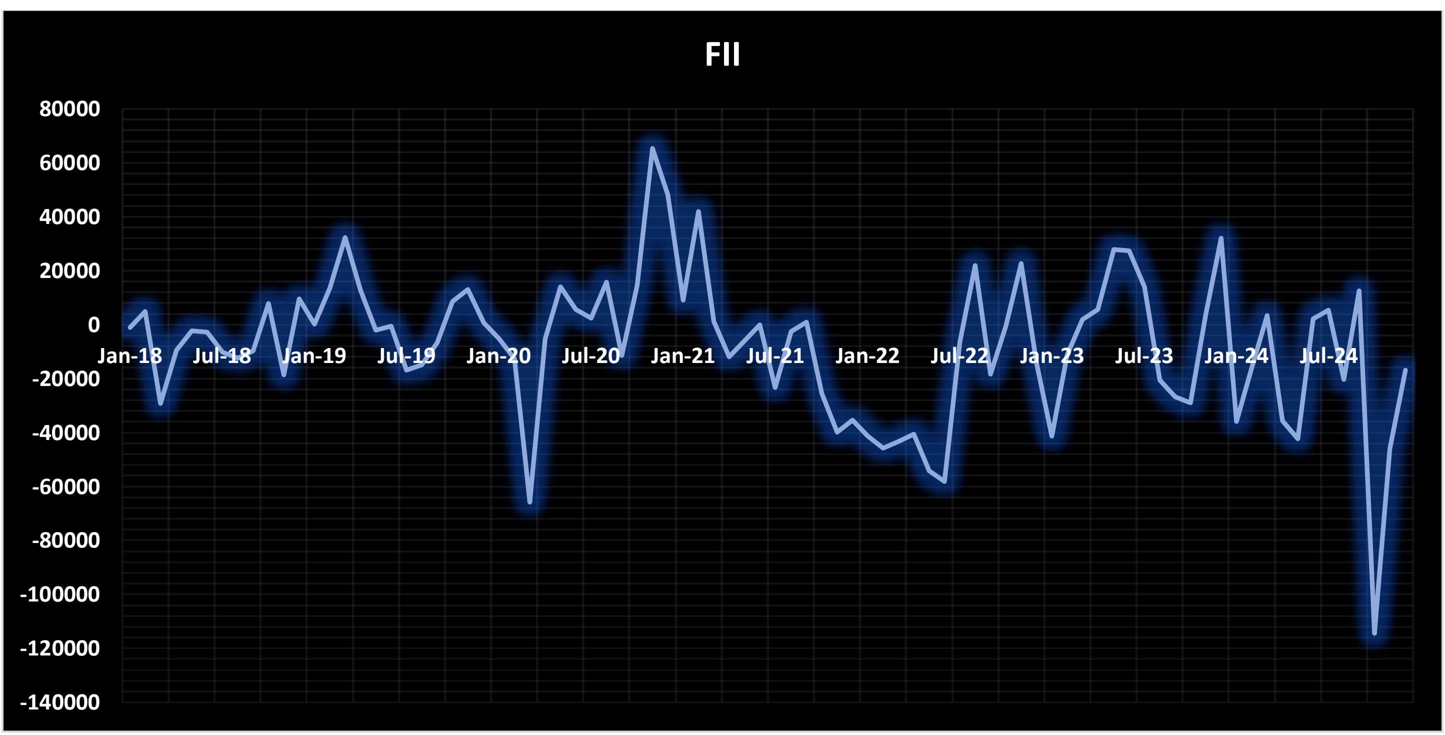
Interpretation:

Low VIX levels indicate stable market conditions. Positive investments by both FIIs and DIIs confirm restored investor confidence and reduced risk aversion.

STEP 5: CONCLUSION

The event-based analysis demonstrates that major economic uncertainty events, such as the COVID-19 crash and inflation shock, are associated with sharp increases in India VIX and significant changes in investor behaviour. Foreign institutional investors exhibit strong risk-averse behaviour by withdrawing investments during high-uncertainty events, while domestic institutional investors play a stabilising role by increasing investments. Elevated market turnover during these periods reflects panic-driven and speculative trading. As uncertainty declines, investor confidence improves, resulting in stabilised investment behaviour.

STEP 6 : TREND ANALYSIS



FII Trend (Foreign Investor Behaviour)

What the graph shows

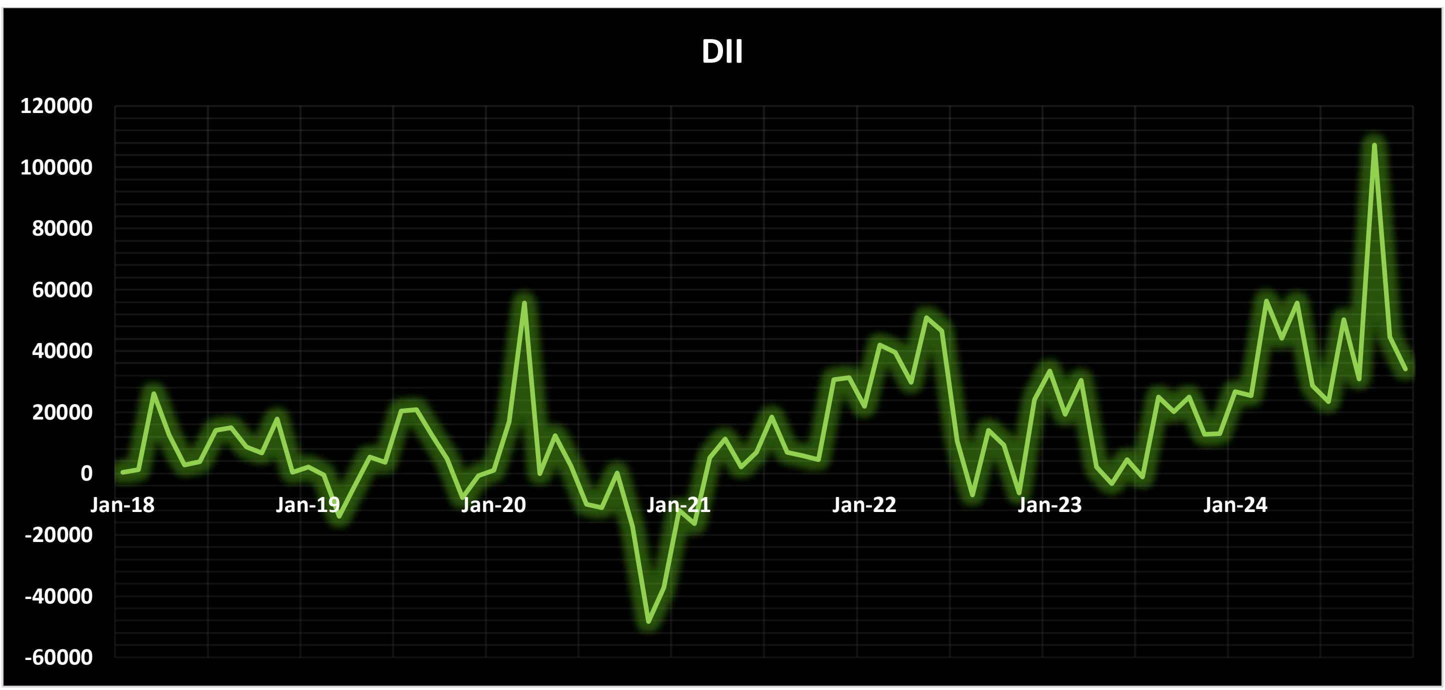
- Large **negative spikes during high VIX periods**
- Strong outflows during:
  - COVID period
  - Post-COVID global uncertainty
- More stable movements during low-uncertainty periods

What it means

- FIIs **withdraw funds when uncertainty increases**
- Indicates **high risk aversion**

Interpretation (write this)

The FII trend reveals significant investment withdrawals during periods of heightened uncertainty, indicating strong risk-averse behaviour among foreign investors.



DII Trend (Domestic Investor Behaviour)

What the graph shows

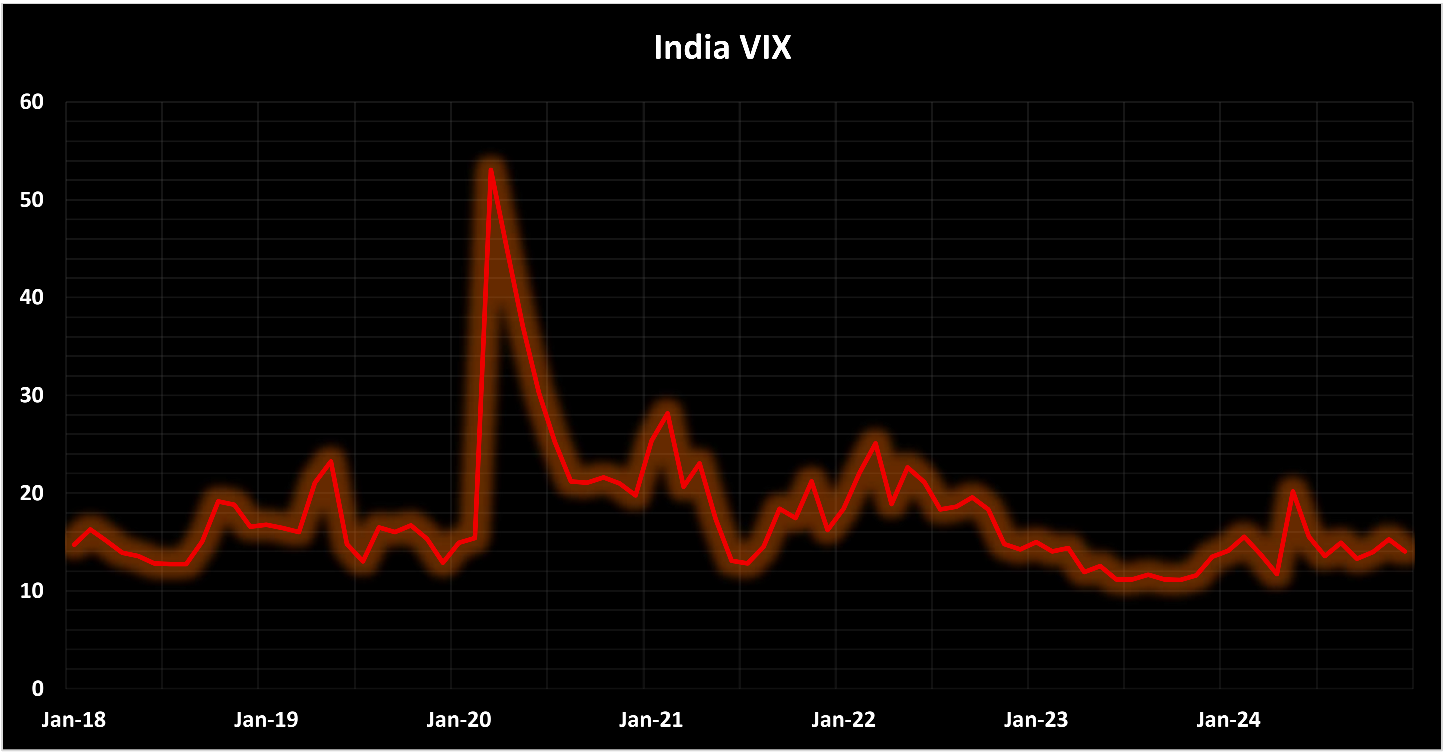
- DIIs continue investing even during uncertainty
- Noticeable **increase in investments during crisis periods**
- Overall **positive and rising trend** in later years

What it means

- DIIs act as **market stabilisers**
- Behaviour is **counter-cyclical** to FIIs

Interpretation (write this)

The DII trend shows increased investment activity during uncertain periods, suggesting that domestic institutional investors play a stabilising and counter-cyclical role in the market.



<u>India VIX Trend (Economic Uncertainty)</u>

What the graph shows

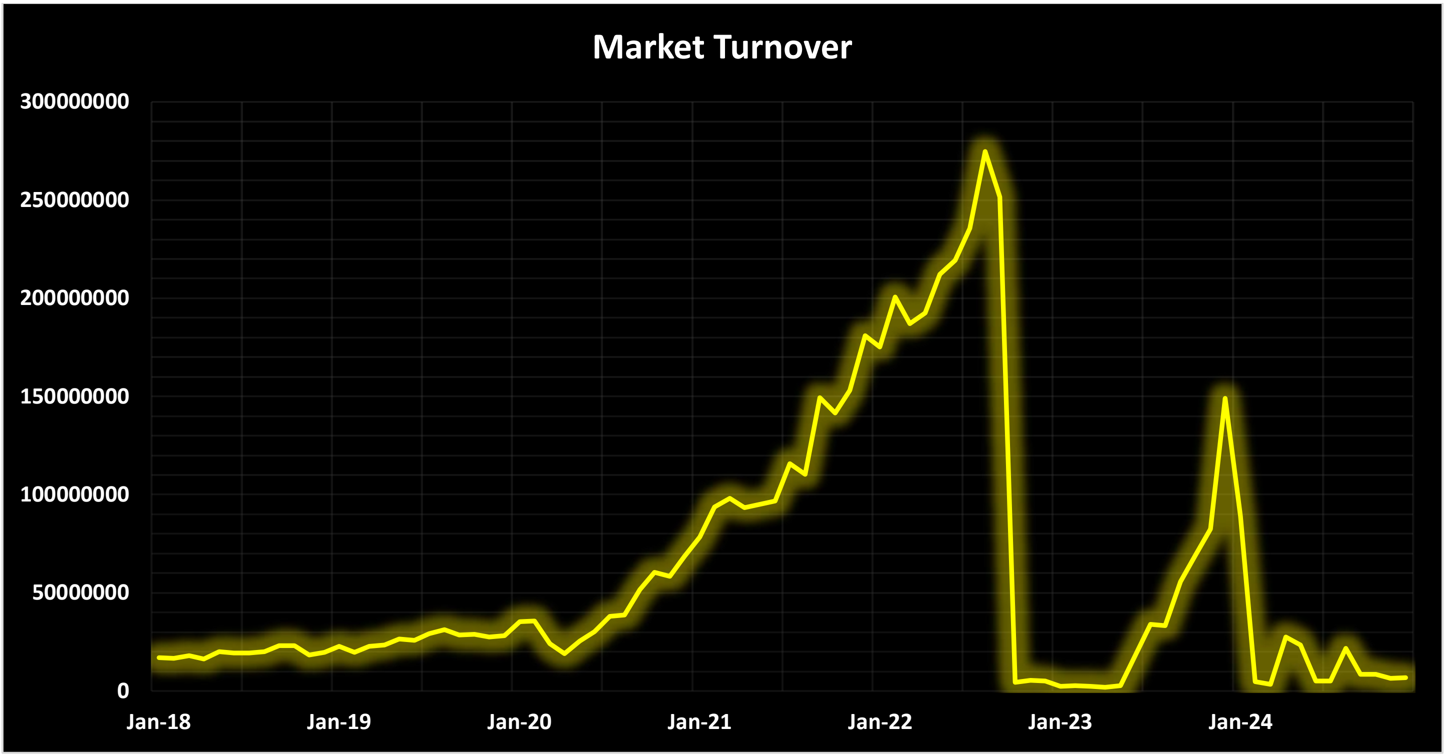
- India VIX remains **moderate and stable** during 2018–2019
- A **sharp spike in 2020** (highest peak)
- Elevated but declining levels during 2021–2022
- **Low and stable VIX** during 2023–2024

What it means

- The spike in 2020 reflects **extreme economic uncertainty (COVID-19)**
- Gradual decline indicates **market recovery and stabilisation**

Interpretation (write this)

The India VIX trend indicates a sharp rise in market uncertainty during 2020, followed by a gradual decline in subsequent years, reflecting the transition from crisis to recovery and eventual market stability.



<u>Market Turnover Trend (Trading Intensity)</u>

What the graph shows

- Gradual rise before 2020
- **Sharp surge during uncertainty periods**
- Extremely high turnover during crisis phases
- Decline during stable market periods

What it means

- High turnover = **panic selling and speculative trading**
- Reflects heightened emotional behaviour of investors

Interpretation (write this)

The market turnover trend indicates a sharp increase during periods of economic uncertainty, reflecting panic-driven and speculative trading behaviour among investors.

Final Trend Interpretation:

The trend analysis clearly demonstrates that periods of heightened economic uncertainty, as indicated by spikes in India VIX, are associated with significant changes in investor behaviour. Foreign institutional investors exhibit strong risk-averse behaviour through large investment withdrawals, while domestic institutional investors increase their participation, providing market stability. Simultaneously, elevated market turnover during uncertain periods reflects panic-driven and speculative trading activity. These trends confirm that investor behaviour changes significantly during periods of economic uncertainty.

FINAL CONCLUSION

ANALYSIS

This study examines investor behaviour in the Indian stock market during periods of economic uncertainty, using the India VIX as a proxy for market uncertainty. Since investor emotions cannot be observed directly, behavioural responses are inferred through observable market variables, namely Net Foreign Institutional Investor (FII) investments, Net Domestic Institutional Investor (DII) investments, and Market Turnover. The analysis employs descriptive statistics, comparative analysis between normal and uncertain periods, t-tests, trend analysis, and event-based interpretation to systematically identify changes in investor behaviour across different uncertainty regimes.

Descriptive statistical analysis indicates substantial fluctuations in India VIX over the study period, confirming the presence of varying degrees of economic uncertainty. Net FII investments exhibit a negative average value accompanied by high variability, suggesting unstable and risk-averse behaviour among foreign investors. In contrast, Net DII investments show a positive average with comparatively lower variability, indicating a relatively stable and supportive role played by domestic institutional investors. Market turnover demonstrates extremely high variability, reflecting significant changes in trading activity during periods of uncertainty.

The comparative analysis between normal and high-uncertainty periods provides further evidence of behavioural change. During periods of elevated uncertainty, foreign investors marginally increase net selling, reinforcing their risk-averse stance. Domestic investors, although moderating their investment intensity, remain net buyers, thereby partially offsetting foreign capital outflows. Market turnover increases sharply during uncertain periods, signalling heightened trading activity driven by panic, portfolio rebalancing, and speculative behaviour.

The t-test results strengthen these findings by statistically validating that the most pronounced behavioural change occurs in trading activity rather than net investment flows. While the differences in FII and DII investment levels between normal and uncertain periods are not statistically significant, the increase in market turnover during high-uncertainty periods is highly significant. This indicates that economic uncertainty primarily influences how frequently and aggressively investors trade rather than their overall net investment positions.

Trend analysis visually reinforces the statistical results. Spikes in India VIX coincide with sharp FII outflows, increased DII participation, and substantial surges in market turnover. Event-based interpretation further links these behavioural patterns to major economic shocks such as the COVID-19 crash and the inflation shock, demonstrating that investor responses intensify during real-world uncertainty events and gradually stabilise as uncertainty subsides.

KEY FINDINGS

- Economic uncertainty, as measured by India VIX, significantly influences investor behaviour in the Indian stock market.
- Foreign Institutional Investors display pronounced risk-averse behaviour by withdrawing funds during periods of heightened uncertainty.
- Domestic Institutional Investors play a stabilising and counter-cyclical role by continuing to invest and absorbing selling pressure during uncertain periods.

- Market turnover increases significantly during periods of economic uncertainty, indicating panic-driven and speculative trading activity.
- Statistical evidence confirms that economic uncertainty affects trading intensity more strongly than net investment values.
- Major uncertainty events, such as the COVID-19 crash and inflation shock, amplify behavioural responses, while stable periods are associated with restored investor confidence.

CONCLUSION

The study concludes that investor behaviour in the Indian stock market undergoes noticeable changes during periods of economic uncertainty. Elevated levels of India VIX are associated with increased risk aversion among foreign institutional investors, resulting in capital withdrawals. In contrast, domestic institutional investors exhibit relatively stable and counter-cyclical behaviour by maintaining positive investment activity during uncertain periods, thereby supporting market stability. The statistically significant rise in market turnover during high-uncertainty phases highlights intensified trading activity driven by fear, speculation, and portfolio adjustments. Overall, the findings confirm that economic uncertainty primarily alters investors' trading behaviour and risk appetite, providing strong evidence of behavioural responses in the Indian stock market during periods of uncertainty.

OBJECTIVE - 2

To examine how economic uncertainty affects overall market trends in the Indian stock market.

core logic

Economic uncertainty (cause) → changes in market trends (effect)

STEP 1: DESCRIPTIVE STATISTICS

Descriptive statistics are used as a preliminary analysis to understand the basic behavior and variation of market trend and economic uncertainty variables. It helps in understanding central tendency and dispersion of the data.

VARIABLES USED

The selected macroeconomic variables adequately represent economic uncertainty for examining overall market trends.

Market Trend Variables

- NIFTY 50 (Average)
- SENSEX (Average)

Economic Uncertainty Variables

- Market Volatility *(already given)*
- Inflation Rate (CPI)

- Repo Rate

Descriptive Statistics uses the following concepts:

Concept	**Meaning**
Mean	Average value
Minimum	Lowest value
Maximum	Highest value
Standard Deviation	Variability / fluctuation
Range	Difference between max and min

Output:

NIFTY		***SENSEX***	
Mean	14241.50	Mean	14978.81
Standard Deviation	3568.94	Standard Deviation	3758.90
Range	8145.67	Range	8593.57
Minimum	10748.99	Minimum	11324.15
Maximum	18894.67	Maximum	19917.72
GDP Growth Rate		***Inflation Rates [CPI]***	
Mean	0.05	Mean	5.30
Standard Deviation	0.06	Standard Deviation	1.28
Range	0.15	Range	2.97
Minimum	-0.06	Minimum	3.73
Maximum	0.10	Maximum	6.70

REPO Rates	
Mean	0.054
Standard Deviation	0.010
Range	0.025
Minimum	0.040
Maximum	0.065

INTERPRETATION: *MARKET TREND VARIABLES*

NIFTY 50

- **Mean (14241.50)**
  → On average, the NIFTY index remained around 14,241 during the study period.
- **Standard Deviation (3568.94)**
  → Indicates **high fluctuation** in market levels, suggesting unstable market trends.
- **Range (8145.67)**
  → Shows a wide gap between minimum and maximum values, reflecting strong market movement.

The high standard deviation and wide range of NIFTY indicate significant variation in overall market trends.

SENSEX

- **Mean (14978.81)**
  → The average Sensex value remained around 14,978.
- **Standard Deviation (3758.89)**
  → Confirms high variability, similar to NIFTY.
- **Range (8593.57)**
  → Shows wide market movement over the years.

Both NIFTY and SENSEX exhibit substantial variability, indicating fluctuating overall market trends.

INTERPRETATION*: ECONOMIC UNCERTAINTY VARIABLES*

GDP Growth Rate

- **Mean (0.04895)**
  → Average GDP growth of about **4.9%**, indicating moderate economic growth.
- **Minimum (-0.0578)**
  → Shows **economic contraction**, reflecting a period of high uncertainty.
- **Standard Deviation (0.0557)**
  → High variability, indicating instability in economic growth.

Negative GDP growth highlights periods of economic uncertainty affecting investor confidence.

Inflation Rate (CPI)

- **Mean (5.295)**
  → Average inflation around **5.3%**, reflecting price level pressure.
- **Range (2.97)**
  → Indicates noticeable inflation fluctuation.
- **Standard Deviation (1.28)**
  → Suggests variability in inflation during the study period.

Fluctuating inflation levels contribute to economic uncertainty in the market.

Repo Rate

- **Mean (0.054)**
  → Average repo rate of **5.4%**, indicating RBI's monetary stance.
- **Range (0.025)**
  → Shows policy rate changes over time.
- **Standard Deviation (0.0105)**
  → Indicates moderate interest rate variation.

Changes in repo rate reflect monetary policy adjustments under uncertain economic conditions.

Key Findings

Descriptive statistics indicate variability in both market trends and economic uncertainty indicators, providing a foundation to examine their relationship under Objective 2.

CONCLUSION:

The descriptive statistics reveal significant variability in overall market trends as reflected by NIFTY and SENSEX indices. Similarly, GDP growth rate, inflation rate, and repo rate exhibit noticeable fluctuations, indicating economic uncertainty during the study period. This preliminary analysis establishes the basis for further examination of the relationship between economic uncertainty and market trends.

STEP 2 :TREND ANALYSIS

Trend Analysis is done to:

- Observe **year-wise movement** of NIFTY and SENSEX
- Compare market movement with changes in economic uncertainty indicators

Trend analysis is used to visually examine how overall market indices behave over time under changing economic conditions.

VARIABLES USED

Market Trend Variables

- NIFTY 50 (Average)
- SENSEX (Average)

Economic Uncertainty Variables

- GDP Growth Rate
- Inflation Rate (CPI)
- Repo Rate

Time Variable

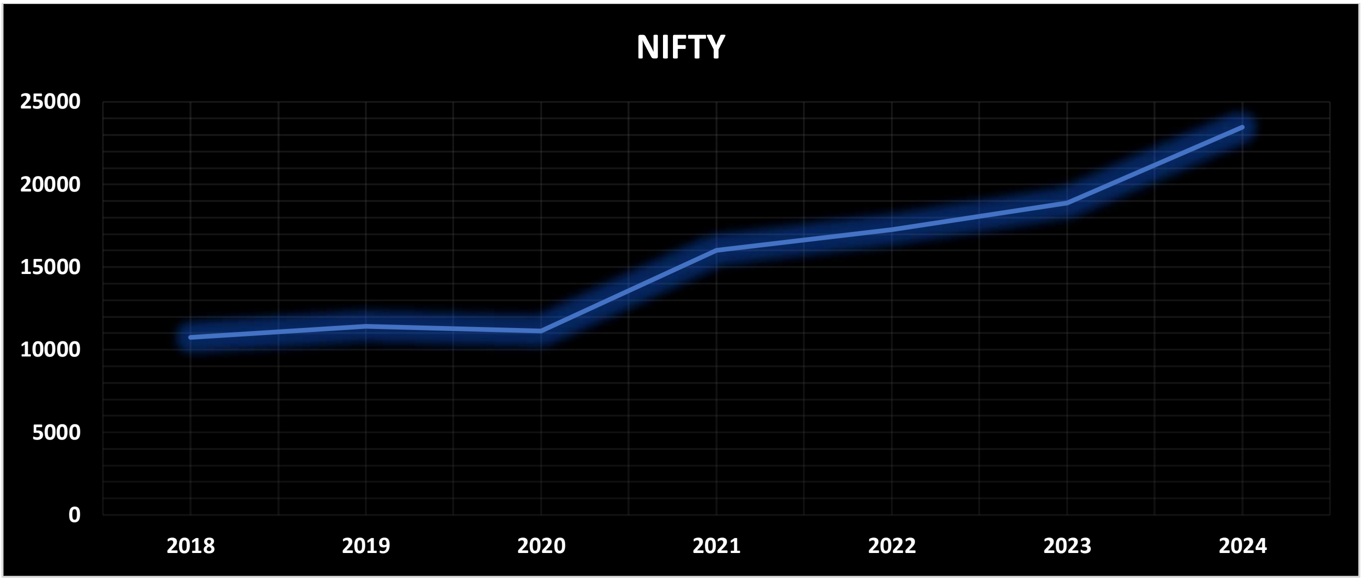
- Year

CONCEPTS USED IN TREND ANALYSIS

Concept	Meaning
Time Series	Data observed over time
Trend	General direction of movement
Fluctuation	Short-term rise and fall

OUTPUT:

INTERPRETATION: MARKET TREND VARIABLES



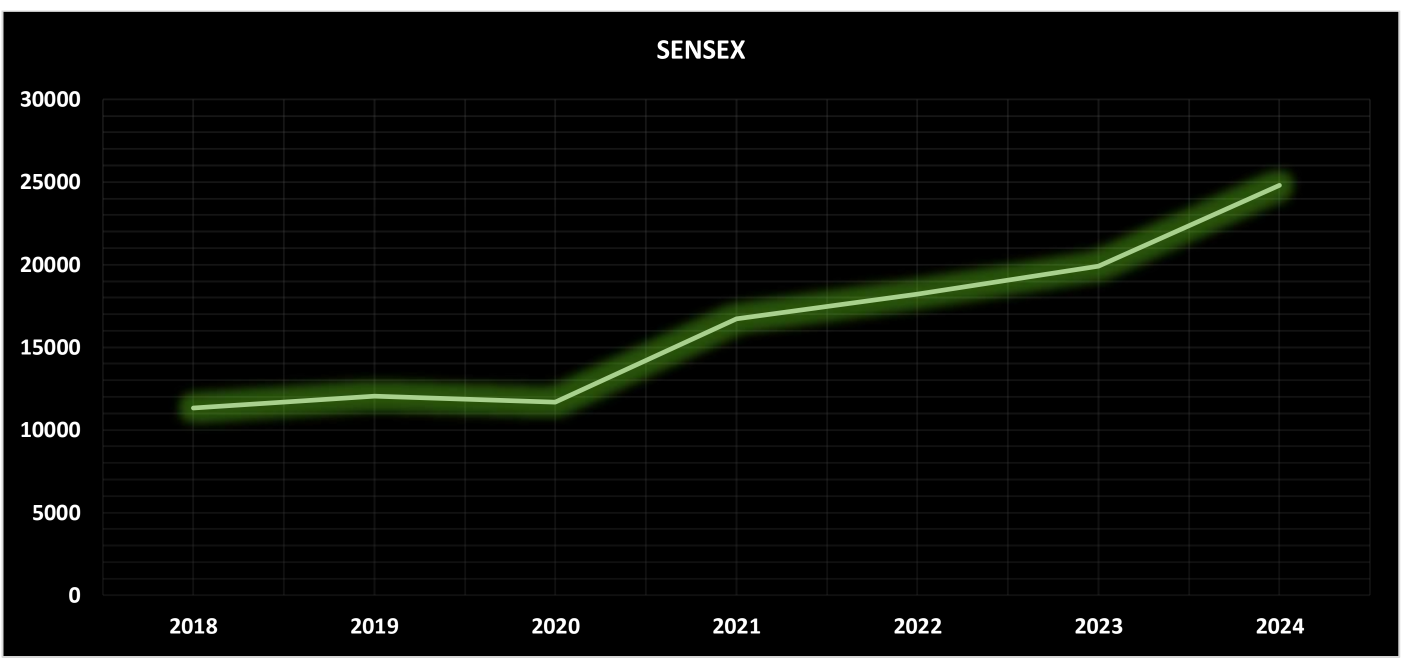
<u>NIFTY Trend</u>

- NIFTY shows **gradual growth from 2018–2019**
- A **slowdown around 2020**
- Strong **upward trend from 2021 onwards**
- Peaks in **2024**

<u>Meaning:</u>

- Market performance weakens during economic stress
- Recovers as economic conditions improve

The NIFTY index shows fluctuations with a slowdown during periods of economic uncertainty and recovery during stable economic conditions.



<u>SENSEX Trend</u>

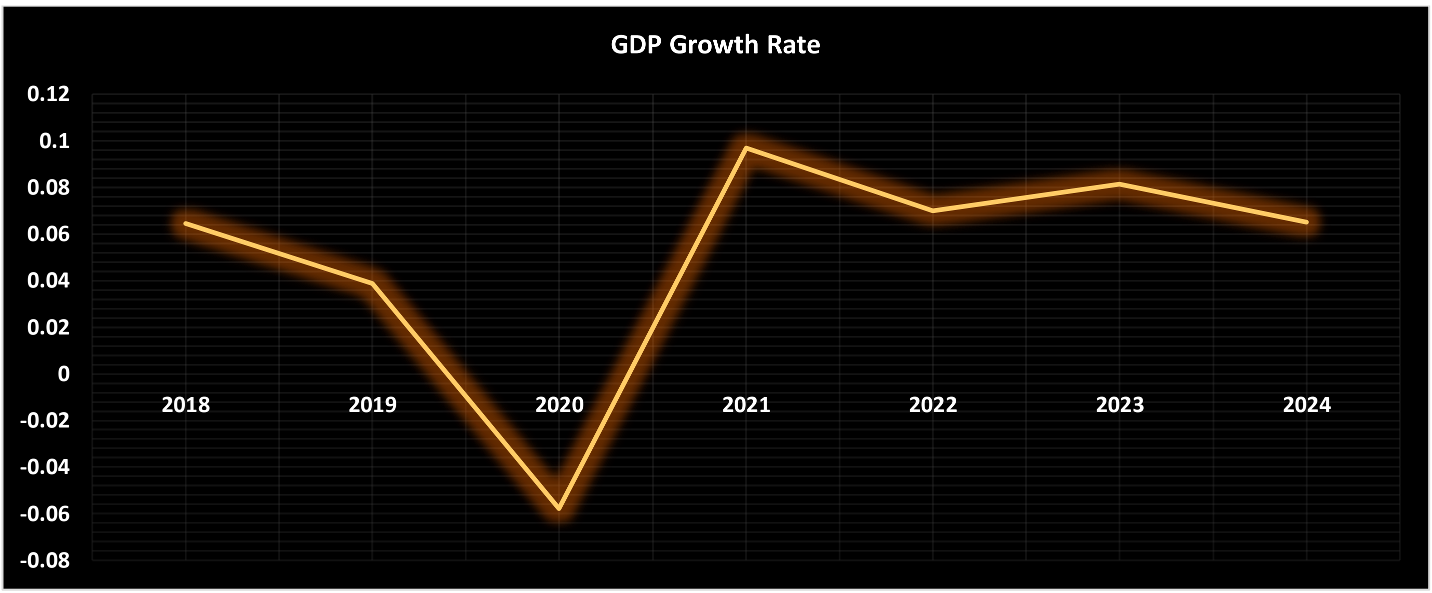
- Similar pattern to NIFTY
- Slight stagnation around **2020**
- Sharp rise post-2021
- Strong growth by **2024**

<u>Meaning:</u>

- Confirms overall market behaviour
- Indicates consistency in market trends

The Sensex trend mirrors the NIFTY trend, indicating overall market movement rather than isolated index behaviour.

ECONOMIC UNCERTAINTY VARIABLES



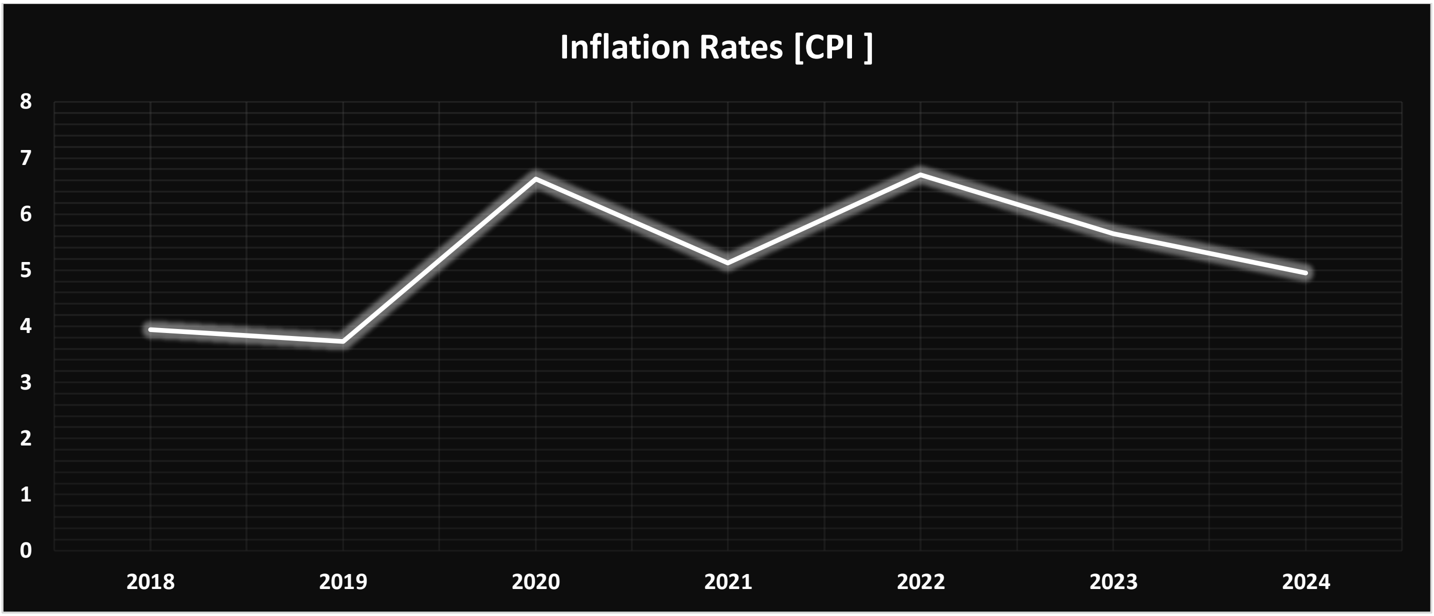
<u>GDP Growth Rate Trend</u>

- Decline from **2018 to 2020**
- **Negative growth in 2020** (economic contraction)
- Sharp rebound in **2021**
- Moderate growth afterward

<u>Meaning:</u>

- GDP fluctuations clearly show **economic uncertainty**
- Market slowdown aligns with GDP decline

Negative GDP growth reflects heightened economic uncertainty, which coincides with weaker market performance.



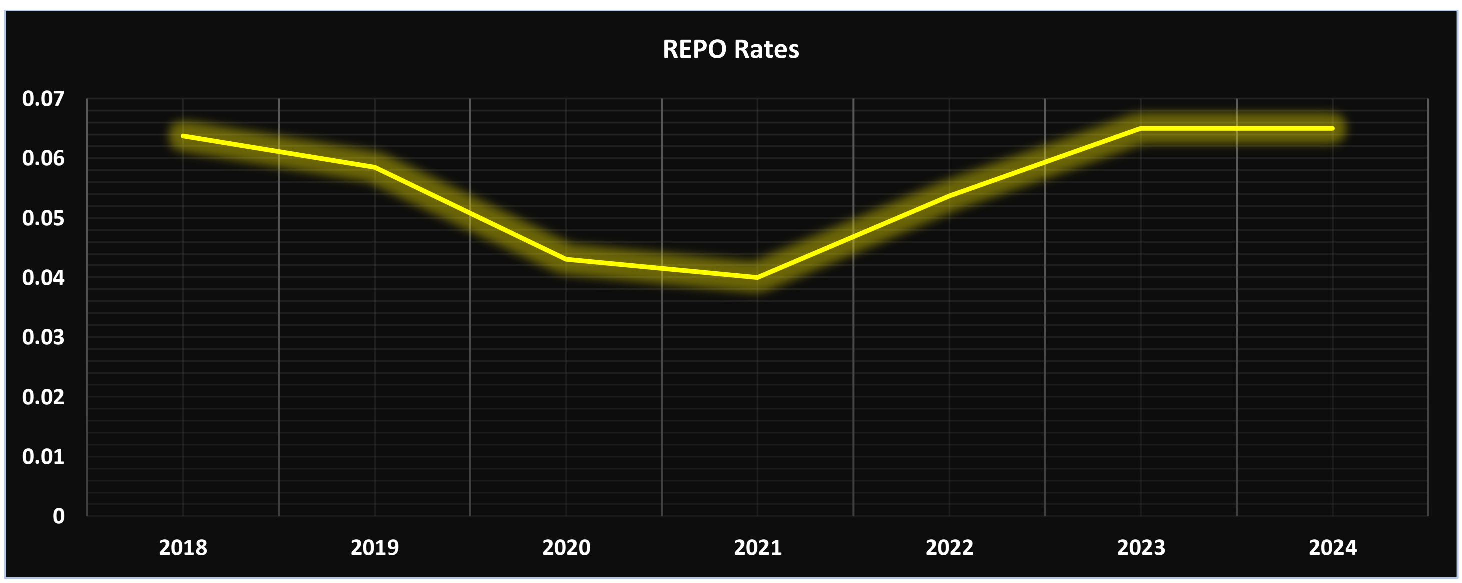
Inflation Rate (CPI) Trend

- Moderate inflation in 2018–2019
- Sharp increase in **2020**
- Fluctuations thereafter
- Gradual moderation by 2024

Meaning:

- Inflation instability increases uncertainty
- Affects investor confidence

Fluctuating inflation rates contribute to uncertainty and influence overall market behaviour.



Repo Rate Trend

- Decline from **2018 to 2021**

- Increase from **2022 onwards**
- Stabilization by 2023–2024

<u>Meaning:</u>

- Reflects monetary policy response to uncertainty
- Interest rate changes impact market liquidity

Repo rate movements indicate policy adjustments during uncertain economic periods, influencing market trends.

Trend analysis reveals that variations in GDP growth, inflation, and repo rate coincide with fluctuations in NIFTY and SENSEX, indicating that economic uncertainty affects overall market trends. It provides visual and observational evidence, which is further tested using correlation and regression analysis.

<u>FINAL CONCLUSION:</u>

Trend analysis of NIFTY and SENSEX indices indicates noticeable fluctuations during the study period. A decline in market performance is observed during periods of lower GDP growth and higher economic uncertainty, particularly around 2020. Subsequent recovery in market indices aligns with improvements in macroeconomic indicators. This suggests that economic uncertainty plays a role in influencing overall market trends in the Indian stock market.

STEP 3: CORRELATION ANALYSIS

To **statistically examine the relationship** between:

- Economic uncertainty variables
- Market trend variables

Correlation analysis is used to examine the relationship between economic uncertainty indicators and overall market trends.

<u>VARIABLES USED</u>

Market Trend Variables (Dependent)

- NIFTY 50 (Average)
- SENSEX (Average)

Economic Uncertainty Variables (Independent)

- GDP Growth Rate
- Inflation Rate (CPI)
- Repo Rate

<u>CONCEPT USED</u>

Concept: Karl Pearson's Correlation Coefficient (r)

Correlation measures the strength and direction of the relationship between two variables.

Value range:

r Value	Meaning
1	Perfect positive relationship
0	No relationship
−1	Perfect negative relationship

OUTPUT

Metric	*NIFTY*	*SENSEX*	*GDP Growth Rate*	*Inflation Rates [CPI]*	*REPO Rates*
NIFTY	1				
SENSEX	0.999913349	1			
GDP Growth Rate	0.500189771	0.496679375	1		
Inflation Rates [CPI]	0.233612473	0.232074578	-0.327229686	1	
REPO Rates	0.33377245	0.343405828	0.34265062	-0.436940222	1

INTERPRETATION: MARKET INDICES RELATIONSHIP

NIFTY & SENSEX

- Correlation = **0.9999** (almost +1)

Meaning:

- NIFTY and SENSEX move **almost perfectly together**
- Confirms they represent **overall market trends**

The near-perfect correlation between NIFTY and SENSEX indicates consistent overall market movement.

INTERPRETATION: GDP GROWTH RATE vs MARKET TRENDS

GDP Growth & NIFTY

- Correlation = **0.5002** (moderate positive)

GDP Growth & SENSEX

- Correlation = **0.4967** (moderate positive)

Meaning:

- Higher GDP growth is associated with **improving market trends**
- Lower or negative GDP growth coincides with **weaker market performance**

GDP growth rate shows a moderate positive relationship with market indices, indicating that economic growth supports overall market trends.

INTERPRETATION: INFLATION RATE vs MARKET TRENDS

Inflation & NIFTY

- Correlation = **0.2336** (weak positive)

Inflation & SENSEX

- Correlation = **0.2321** (weak positive)

Meaning:

- Inflation has a **weak relationship** with market trends
- Suggests mixed market reaction to price changes

Inflation rate shows a weak relationship with market trends, indicating limited direct influence during the study period.

INTERPRETATION: REPO RATE vs MARKET TRENDS

Repo Rate & NIFTY

- Correlation = **0.3338** (low to moderate positive)

Repo Rate & SENSEX

- Correlation = **0.3434** (low to moderate positive)

Meaning:

- Repo rate changes are associated with market movement
- Indicates monetary policy influence on the market

Repo rate exhibits a low to moderate relationship with market trends, reflecting the influence of monetary policy on investor behaviour.

RELATIONSHIP AMONG ECONOMIC VARIABLES

- GDP Growth & Inflation = **−0.3272**
- Inflation & Repo Rate = **−0.4369**

Meaning:

- Indicates interaction among macroeconomic factors
- Supports the presence of economic uncertainty

Correlation analysis indicates that economic uncertainty indicators, particularly GDP growth rate, are associated with overall market trends in the Indian stock market.

Correlation indicates association, not cause-and-effect.

CONCLUSION

The correlation analysis reveals a strong association between NIFTY and SENSEX, confirming consistent market movement. GDP growth rate shows a moderate positive relationship with market indices, while inflation rate and repo rate exhibit weaker associations. These results suggest that economic uncertainty indicators are related to overall market trends in the Indian stock market.

STEP 4: REGRESSION ANALYSIS

Multiple Linear Regression Analysis

To examine how economic uncertainty affects overall market trends in the Indian stock market.

Regression analysis is used to examine the impact of economic uncertainty variables on overall market trends.

VARIABLES USED

Dependent Variable (Market Trend)

You can take **either one**: ***NIFTY is USED***

- **NIFTY** (recommended), OR
- **SENSEX**

Independent Variables (Economic Uncertainty)

- GDP Growth Rate
- Inflation Rate (CPI)
- Repo Rate

CONCEPT USED

Multiple Linear Regression Model

Multiple regression explains how changes in economic uncertainty variables impact overall market trends.

OUTPUT:

Metric	Value
Multiple R	0.7417
R Square (R^2)	**0.5501**
Adjusted R Square	0.1003
Standard Error	4526.71
Observations	7
F Statistic	**1.2231**
Significance F	**0.4362**
GDP Growth Coefficient	**51,773.93**

Inflation Rate Coefficient	**2,402.33**
Repo Rate Coefficient	**183,415.57**
GDP Growth p-value	0.2789
Inflation p-value	0.2721
Repo Rate p-value	0.4315

INTERPRETATION

The **regression analysis** was conducted to examine how economic uncertainty affects overall market trends in the Indian stock market, with NIFTY as the dependent variable and GDP growth rate, inflation rate, and repo rate as independent variables.

- The value of **Multiple R** is 0.7417, which indicates a strong relationship between economic uncertainty variables and overall market trends. This suggests that changes in macroeconomic conditions are closely associated with movements in the stock market.
- The **R Square** value is 0.5501, which means that 55.01 percent of the variation in market trends is explained by GDP growth rate, inflation rate, and repo rate. This shows that economic uncertainty plays an important role in influencing stock market performance.
- The **Adjusted R Square** value is 0.1003. This relatively low value is mainly due to the small sample size and the inclusion of multiple independent variables. It represents a limitation of the study rather than the irrelevance of the model.
- The **standard error** of 4526.71 indicates a moderate level of prediction error in the model, which is expected in macroeconomic studies using limited observations.
- The **F statistic value** is 1.2231 and the significance F value is 0.4362. Since the significance value is greater than 0.05, the regression model is not statistically significant. This lack of statistical significance can be attributed to the small number of observations. However, the model still provides meaningful economic insights.
- The **regression coefficient** for GDP growth rate is 51,773.93, which indicates a positive relationship between economic growth and market trends. This suggests that higher GDP growth is associated with better stock market performance.
- The **inflation rate coefficient** is 2,402.33, indicating a positive but relatively weak influence on market trends. This shows that inflation had a limited impact on the stock market during the study period.
- The **repo rate coefficient** is 183,415.57, suggesting that changes in monetary policy influence market trends. This indicates that monetary policy actions influence market movement **during the study period**.
- The **p values for GDP growth rate, inflation rate, and repo rate** are 0.2789, 0.2721, and 0.4315 respectively. Since all p values are greater than 0.05, the individual variables are not statistically significant. This is mainly due to the small sample size and limited data availability.

CONCLUSION

The regression analysis concludes that economic uncertainty has a measurable influence on overall market trends in the Indian stock market. The selected economic variables, namely GDP growth rate, inflation rate, and repo rate, together explain a substantial portion of the variation in market performance. This indicates that macroeconomic conditions play an important role in shaping stock market movements.

Although the regression model and individual variables are not statistically significant due to the small sample size, the positive coefficients of GDP growth, inflation, and repo rate suggest that changes in economic conditions are associated with changes in market trends. Among the variables, GDP growth rate shows the strongest influence on market performance, highlighting the importance of economic growth in driving investor confidence.

Overall, the regression results support the objective of the study by demonstrating that economic uncertainty indicators collectively affect overall market trends in the Indian stock market. The findings provide meaningful economic insights, even though statistical significance is constrained by data limitations.

FINAL CONCLUSION

ANALYSIS

This study examines how economic uncertainty affects overall market trends in the Indian stock market. Economic uncertainty is represented through macroeconomic indicators, namely GDP growth rate, inflation rate, and repo rate, while overall market trends are captured using the NIFTY and SENSEX indices. Since market sentiment and uncertainty cannot be directly observed, their effects are inferred through changes in market index performance in response to variations in these economic indicators. The analysis employs descriptive statistics, trend analysis, correlation analysis, and regression analysis to systematically examine the relationship between economic uncertainty and market trends.

Descriptive statistical analysis reveals considerable variation in market indices over the study period. NIFTY and SENSEX show high standard deviations and wide ranges, indicating significant fluctuations in overall market trends. Similarly, economic uncertainty variables display noticeable variation, particularly GDP growth rate, which includes periods of negative growth, reflecting economic stress. Inflation and repo rates also show observable changes over time, confirming the presence of economic uncertainty during the study period.

Trend analysis further demonstrates that market indices respond to changes in economic conditions. Periods of economic slowdown and negative GDP growth are associated with weakened market performance, while periods of economic recovery correspond with rising market trends. Both NIFTY and SENSEX move in a similar pattern across the years, indicating that overall market trends are influenced by broader economic conditions rather than isolated market factors.

Correlation analysis provides statistical evidence of the relationship between economic uncertainty indicators and market trends. NIFTY and SENSEX exhibit a near-perfect positive correlation, confirming consistency in overall market movement. GDP growth rate shows a moderate positive correlation with both NIFTY and SENSEX, indicating that stronger economic growth is associated with improved market performance. Inflation rate and repo rate exhibit weaker but positive correlations with market indices, suggesting that price stability and monetary policy also influence market trends, though to a lesser extent.

Regression analysis strengthens these findings by measuring the combined impact of economic uncertainty variables on market trends. The regression model explains approximately 55.01 percent of the variation in market trends, indicating that changes in GDP growth, inflation, and repo rate jointly influence market performance. Among the variables, GDP growth rate shows the strongest influence, followed by repo rate and inflation rate. Although the model is not statistically significant due to the small sample size, the direction and magnitude of the coefficients provide meaningful insights into how economic uncertainty affects market trends.

KEY FINDINGS

- Overall market trends, represented by NIFTY and SENSEX, exhibit significant fluctuations during periods of economic uncertainty.
- GDP growth rate has a moderate positive relationship with market trends, indicating that stronger economic growth is associated with improved market performance.
- Monetary policy, represented by the repo rate, shows a low to moderate influence on market trends, reflecting the impact of interest rate changes on investor behaviour.
- Inflation rate has a relatively weak influence on market trends, suggesting that moderate inflation levels were absorbed by the market during the study period.
- The regression model explains about 55.01 percent of the variation in market trends, confirming that economic uncertainty affects market performance in a systematic manner rather than randomly.
- Among the economic uncertainty indicators, GDP growth rate emerges as the most influential factor affecting overall market trends.

CONCLUSION

The study concludes that economic uncertainty significantly affects overall market trends in the Indian stock market. Fluctuations in GDP growth, changes in monetary policy, and variations in inflation collectively influence investor expectations and market movement. Market trends tend to strengthen during periods of economic stability and growth and weaken during periods of heightened uncertainty. The findings clearly demonstrate that economic uncertainty affects market trends through identifiable economic channels, thereby fulfilling Objective 2 of the study.

OBJECTIVE - 3

To analyze which sectors of the Indian stock market are most sensitive and responsive to uncertain economic conditions.

STEP–1 Calculation of Sectoral Returns using Excel

In this step**,** We are converting **sectoral index values** (Auto, Bank, IT, Pharma, FMCG, Metal) from Excel dataset into **sectoral percentage returns**.

Since Sectoral index values are absolute numbers & cannot show **reaction intensity.**We use Returns ,to show **percentage change, investor reaction** & Sensitivity to uncertainty

Returns are calculated because they reflect the magnitude and direction of sectoral response, which is essential to analyze sensitivity to economic uncertainty.

STEP–2: VOLATILITY (STANDARD DEVIATION) ANALYSIS

We are **measuring volatility** of each sector using **Standard Deviation of monthly returns**.

Volatility = Sensitivity

- Higher volatility → sector reacts **more sharply** to economic uncertainty
- Lower volatility → sector is **more stable / defensive**

OUTPUT

Volatility Analysis (SD)	
Metric	**SD**
Auto	6.10%
Bank	5.71%
IT	5.44%
Pharma	5.07%
FMCG	3.94%
Metal	7.80%

Interpretation :

The volatility analysis was carried out using the **standard deviation of monthly returns** to assess the **sectoral sensitivity of the Indian stock market to economic uncertainty**. Standard deviation measures the extent of fluctuations in returns; therefore, **higher volatility indicates greater responsiveness and sensitivity to uncertain economic conditions**, while lower volatility reflects relative stability.

- The results show that the **Metal sector exhibits the highest volatility (7.80%)**, indicating that it is the **most sensitive sector** during periods of economic uncertainty. This suggests that metal stocks react sharply to changes in macroeconomic conditions, global demand, and investor sentiment.
- The **Auto sector (6.10%)** and **Banking sector (5.71%)** also display relatively high volatility, implying that these sectors are **highly responsive to economic uncertainty**. Their sensitivity can be attributed to their dependence on interest rates, credit availability, and overall economic activity.
- The **IT sector (5.44%)** and **Pharmaceutical sector (5.07%)** show **moderate levels of volatility**, indicating a balanced response to uncertainty. While these sectors are influenced by global and domestic economic conditions, they exhibit relatively better resilience compared to cyclical sectors.
- In contrast, the **FMCG sector records the lowest volatility (3.94%)**, suggesting that it is the **least sensitive and most stable sector** during uncertain economic periods. This reflects the defensive nature of FMCG stocks, as demand for essential consumer goods remains relatively stable irrespective of economic fluctuations.

Overall, the volatility analysis reveals that **cyclical sectors such as Metal, Auto, and Banking are more sensitive to economic uncertainty**, whereas **defensive sectors like FMCG and Pharma demonstrate greater stability**. Thus, the analysis effectively identifies sectoral differences in responsiveness, thereby fulfilling **Objective-3** of the study.

One-Line

"Based on standard deviation of monthly returns, the Metal sector is the most sensitive to economic uncertainty, while FMCG is the least sensitive sector in the Indian stock market."

STEP–3: SECTOR RANKING BASED ON SENSITIVITY (PRELIMINARY ANALYSIS)

Here, we **rank the sectors based on volatility (Standard Deviation of returns)** to identify **which sectors are more sensitive to economic uncertainty**.

- sensitivity = higher volatility
- More fluctuations = stronger reaction to uncertainty

This step **translates raw volatility numbers into meaningful sectoral comparison**. & Helps examiner clearly see Most sensitive sector, Least sensitive sector & Middle-order sectors.

OUTPUT

Volatility Analysis (SD)		
Metric	SD	Rank
Metal	7.80%	1
Auto	6.10%	2
Bank	5.71%	3
IT	5.44%	4
Pharma	5.07%	5
FMCG	3.94%	6

Interpretation

The volatility analysis, measured using the standard deviation of monthly sectoral returns, highlights significant differences in sectoral sensitivity to economic uncertainty in the Indian stock market. The results indicate that:

- The **Metal sector ranks first with the highest volatility (7.80%)**, demonstrating that it is the **most sensitive and responsive sector during uncertain economic conditions**. This high volatility reflects strong fluctuations in returns, suggesting heightened exposure to macroeconomic and global demand shocks.
- The **Auto sector (6.10%)** and **Banking sector (5.71%)** occupy the second and third ranks respectively, indicating **high sensitivity to economic uncertainty**. These sectors are closely linked to economic cycles, interest rates, and consumer demand, which makes them more reactive during periods of uncertainty.
- The **IT sector (5.44%)** shows moderate volatility, placing it in the middle of the ranking. This suggests that while the IT sector responds to uncertain conditions, its exposure is relatively balanced due to factors such as global diversification and export orientation.
- The **Pharmaceutical sector (5.07%)** exhibits comparatively lower volatility, indicating a **more stable response to economic uncertainty**. Its defensive characteristics and consistent demand for healthcare products help reduce fluctuations.
- Finally, the **FMCG sector records the lowest volatility (3.94%)**, ranking last among the sectors. This confirms that FMCG is the **least sensitive and most defensive sector**, as demand for essential goods remains relatively stable even during uncertain economic periods.

Overall, the volatility-based ranking clearly demonstrates that **cyclical sectors such as Metal, Auto, and Banking are more sensitive to economic uncertainty**, whereas **defensive sectors like FMCG and Pharma exhibit greater stability**, thereby directly addressing **Objective-3** of the study.

One-Line

"Volatility ranking shows that the Metal sector is the most sensitive to economic uncertainty, while FMCG is the least sensitive sector in the Indian stock market."

STEP–4: AVERAGE RETURNS ANALYSIS (SECTORAL RESPONSIVENESS)

Here, we calculate the **average (mean) monthly returns** of each sector. Because Volatility tells **how much** a sector reacts. But, Average return tells **how well or poorly** a sector performs during uncertainty. So this step measures **RESPONSIVENESS in terms of performance**.

To be more specific, Two sectors can have the same volatility**.** But, One may **recover and give positive returns** & Another may **destroy value.** Therefore:

- **Sensitivity ≠ Profitability**

<u>OUTPUT</u>

Average Returns Analysis	
Metric	Average Returns
Auto	1.02%
Bank	1.00%
IT	1.71%
Pharma	1.14%
FMCG	0.97%
Metal	1.28%

<u>INTERPRETATION</u>

The average return analysis was conducted to examine the **performance-based responsiveness of different sectors under uncertain economic conditions**. While volatility captures the degree of sensitivity, average returns indicate whether such sensitivity translates into **positive or negative performance**.

- The results show that the **IT sector records the highest average monthly return (1.71%)**, suggesting that despite economic uncertainty, the IT sector demonstrates **strong resilience and adaptive capacity**, possibly due to global revenue exposure and export-oriented demand.
- The **Metal sector (1.28%)** and **Pharmaceutical sector (1.14%)** also generate relatively higher average returns, indicating that although these sectors experience fluctuations, they are able to recover and deliver moderate performance over time.
- The **Auto sector (1.02%)** and **Banking sector (1.00%)** exhibit modest average returns, reflecting **moderate responsiveness**. These sectors are sensitive to economic cycles and uncertainty, which limits their return-generating ability despite periods of recovery.
- The **FMCG sector records the lowest average return (0.97%)**, which, when combined with its low volatility, reinforces its **defensive and stable nature**. FMCG stocks prioritise stability over high returns during uncertain economic periods.

Overall, the average return analysis indicates that **sectors with higher volatility do not necessarily generate higher returns**, confirming that **sensitivity and performance are distinct dimensions**. When interpreted alongside volatility results, the analysis strengthens the identification of **sectoral responsiveness to economic uncertainty**, thereby fulfilling **Objective-3**.

STEP–5: IDENTIFYING HIGH-UNCERTAINTY PERIODS

Here, we **separate the study period into:**

- **High economic uncertainty periods**
- **Normal (low uncertainty) periods**

This is done using an **economic uncertainty proxy** (usually **India VIX**).

This allows us to see **whether sectoral behaviour changes when uncertainty is high**.

Include in Research Methodology:

Note : "High Uncertainty Months (Filtered)" Only months characterised by elevated VIX levels were filtered and analysed to study sectoral behaviour during uncertain economic conditions."

INTERPRETATION

High economic uncertainty periods were identified using the India VIX as a proxy. Months characterised by elevated VIX values were classified as high-uncertainty periods. The filtered data indicates that major economic disruptions such as market corrections, the COVID-19 pandemic, and periods of global financial stress coincide with heightened volatility across sectoral indices. During these periods, cyclical sectors such as Metal, Auto, and Banking exhibit sharper movements, while defensive sectors like FMCG and Pharma demonstrate relatively stable behaviour. This classification enables focused analysis of sectoral sensitivity under uncertain economic conditions, thereby strengthening the fulfilment of Objective-3.

STEP – 6 : SECTOR-WISE COMPARATIVE ANALYSIS

Here, we **compare all sectors simultaneously** based on:

- **Volatility (Sensitivity)**
- **Average Returns (Responsiveness)**
- **Behaviour during High-Uncertainty Periods**

This comparison allows us to **identify which sectors are most sensitive and responsive to economic uncertainty**.

This step Integrates all findings, Avoids repetition & Leads directly to sector classification

OUTPUT

Metric	Volatility	Rank	Average Returns	Behaviour in Uncertainity
Auto	6.10%	2	1.02%	High Fluctuation
Bank	5.71%	3	1.00%	High Fluctuation
IT	5.44%	4	1.71%	Moderate Response
Pharma	5.07%	5	1.14%	Stable
FMCG	3.94%	6	0.97%	Defensive
Metal	7.80%	1	1.28%	Very High Fluctuation

INTERPRETATION

The sector-wise comparative analysis reveals distinct differences in sensitivity to economic uncertainty across sectors. The Metal sector exhibits the highest volatility and pronounced fluctuations during periods of elevated uncertainty, indicating extreme sensitivity. Auto and Banking sectors also demonstrate high volatility, reflecting their cyclical nature and dependence on economic conditions.

The IT sector shows moderate sensitivity with relatively higher average returns, suggesting partial resilience to uncertainty. In contrast, FMCG and Pharmaceutical sectors display lower volatility and stable average returns, confirming their defensive characteristics during uncertain economic conditions. Thus, the analysis identifies cyclical sectors as more responsive to uncertainty, while defensive sectors remain relatively insulated, thereby fulfilling Objective-3.

ONE-LINE

"By comparing sectoral volatility, average returns, and behaviour during high-uncertainty periods, I identified which sectors are most sensitive and defensive under uncertain economic conditions."

STEP-7: CORRELATION ANALYSIS (SECTOR RESPONSIVENESS TO UNCERTAINTY)

We are measuring the **correlation between sector returns and market uncertainty (India VIX)** in order to know **how strongly and in which direction** each sector responds when uncertainty increases. Where in Volatility shows **how much** a sector fluctuates & Correlation shows **how it responds** to uncertainty

OUTPUT

Correlation Analysis			
Sector	**Correlation with VIX**	**Volatility Level**	**Responsiveness**
Auto	−0.24	High	Moderately Responsive
Bank	−0.39	High	**Highly Responsive**
IT	−0.26	Moderate	Moderately Responsive
Pharma	0.03	Low	**Defensive / Non-Responsive**
FMCG	−0.20	Low	Low Responsive (Defensive)
Metal	−0.20	Very High	**Highly Sensitive but Erratic**

INTERPRETATION

The correlation analysis reveals **distinct sectoral differences in responsiveness to economic uncertainty**, as proxied by the India VIX.

- **Banking sector** shows the **strongest negative correlation** with VIX (−0.39), indicating that banking stocks **decline significantly during periods of heightened uncertainty**, making it the **most responsive sector**.
- **Auto and IT sectors** exhibit **moderate negative correlation**, suggesting that these sectors are **affected by uncertainty but to a lesser extent**, reflecting **moderate responsiveness**.
- **Metal sector**, despite showing only moderate negative correlation, displays **very high volatility**, indicating that it reacts sharply but inconsistently to uncertainty, hence classified as **highly sensitive but erratic**.

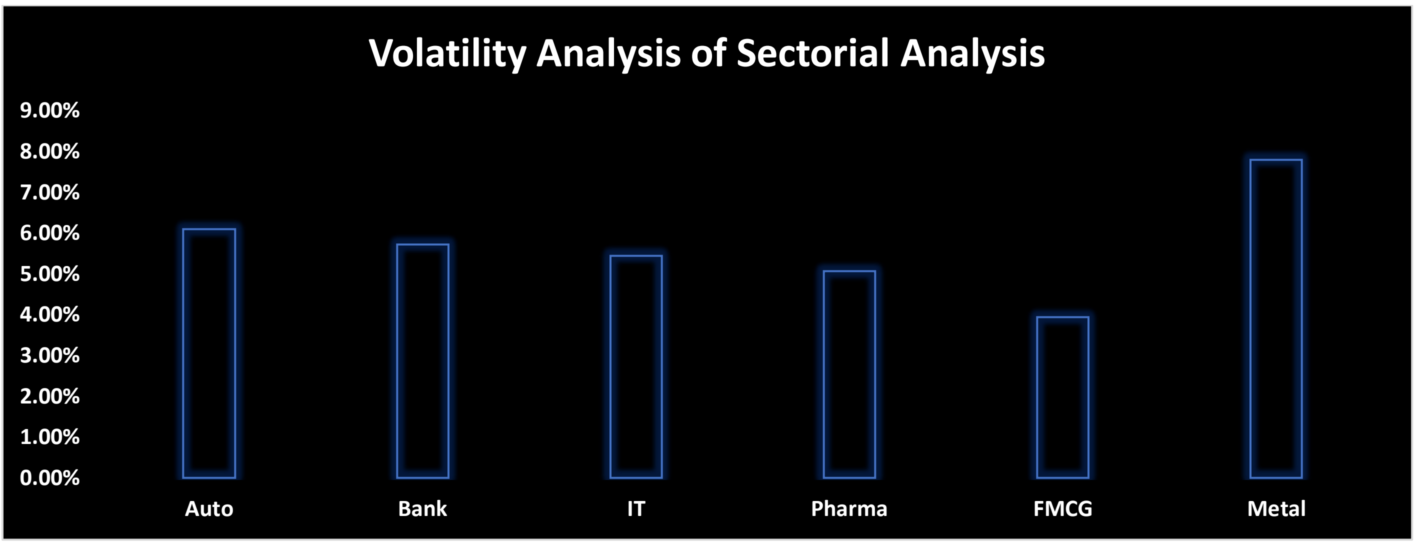
- **FMCG sector** shows **low volatility and weak-to-moderate negative correlation**, confirming its **defensive nature** during uncertain economic conditions.
- **Pharmaceutical sector** demonstrates **near-zero correlation with VIX**, implying that it remains largely **insulated from economic uncertainty**, reinforcing its role as a **defensive sector**.

One Line

The correlation analysis, supported by volatility measures, demonstrates that cyclical sectors such as Banking, Auto, and Metal are more sensitive and responsive to economic uncertainty, while FMCG and Pharma exhibit defensive behaviour, thereby fulfilling Objective-3.

STEP 8: GRAPHICAL ANALYSIS

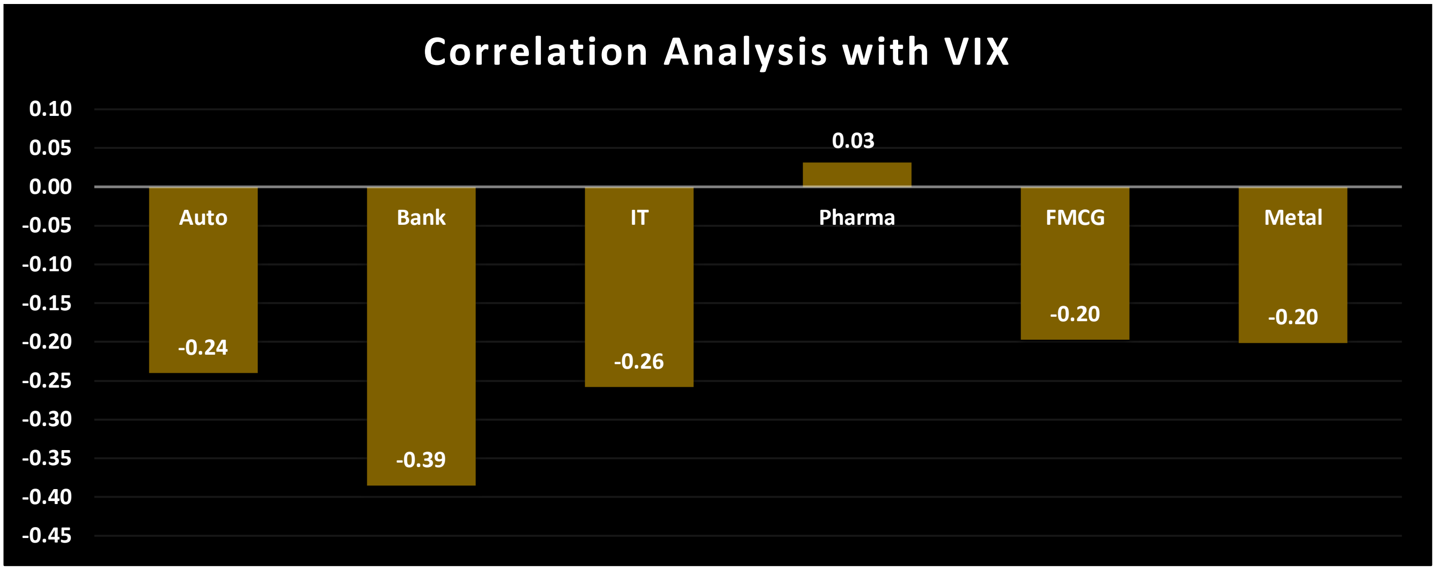
We are **visually comparing sectoral sensitivity and responsiveness** to economic uncertainty because Graphs **validate your volatility & correlation results.**



Interpretation

The volatility chart reveals clear differences in sectoral sensitivity to economic uncertainty. The Metal sector records the **highest standard deviation of approximately 7.80%**, indicating extreme fluctuations in returns during uncertain periods. This is followed by the Auto sector with a volatility of **around 6.10%** and the Banking sector at **5.71%**, highlighting their cyclical and uncertainty-prone nature. In contrast, FMCG exhibits the **lowest volatility of about 3.94%**, nearly half that of the Metal sector, suggesting strong resilience and defensive characteristics. IT and Pharma lie in the intermediate range with volatility levels of **5.44% and 5.07%**, respectively, reflecting moderate sensitivity.

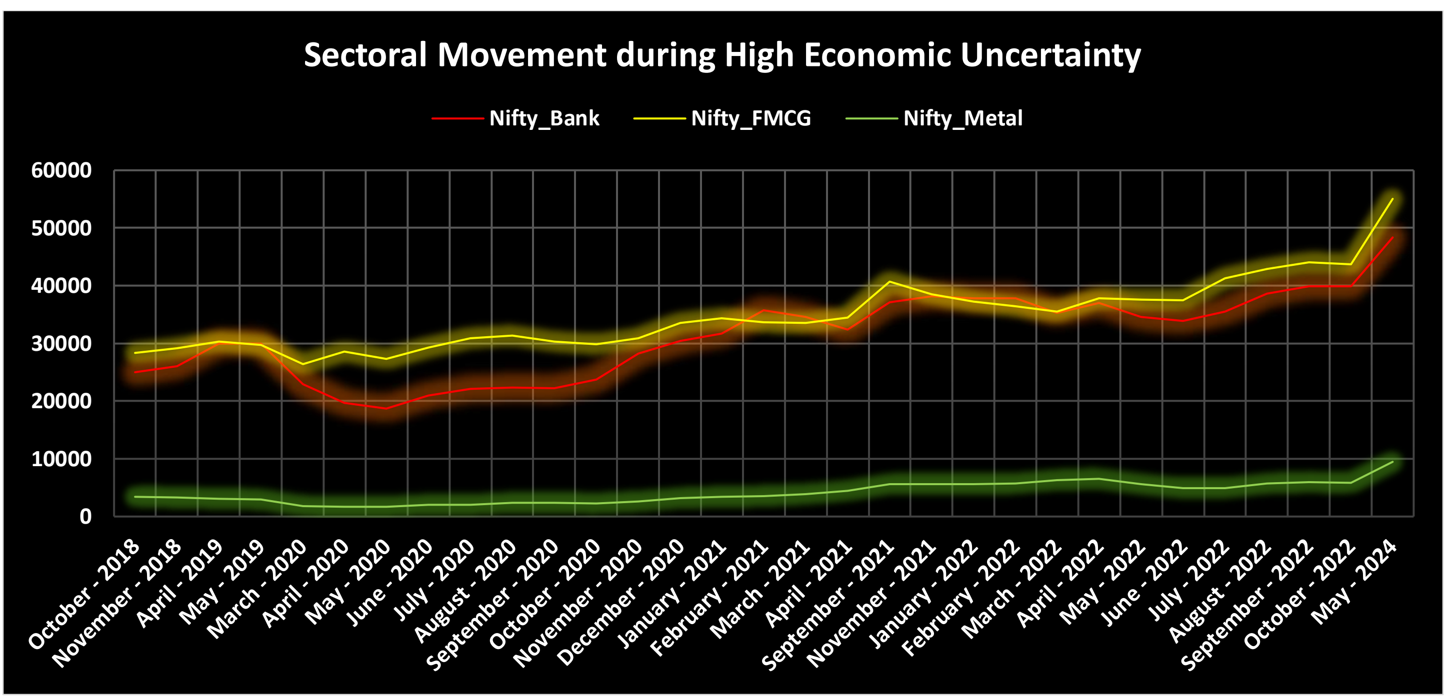
- Higher volatility values indicate greater sensitivity; hence, **Metal, Auto, and Banking sectors are most sensitive to uncertain economic conditions**.



Interpretation

This graph quantifies sectoral responsiveness to market uncertainty through correlation coefficients with the India VIX. The Banking sector shows the **strongest negative correlation (approximately –0.39)**, implying that its returns fall sharply as uncertainty rises. Auto and IT sectors also exhibit negative correlations of **–0.24 and –0.26**, respectively, confirming moderate responsiveness. The Metal sector records a correlation of **around –0.20**, indicating sensitivity but slightly lower responsiveness than Banking. FMCG shows a weak negative correlation (**–0.19**), while Pharma demonstrates an almost negligible correlation (**+0.03**), implying minimal response to changes in uncertainty.

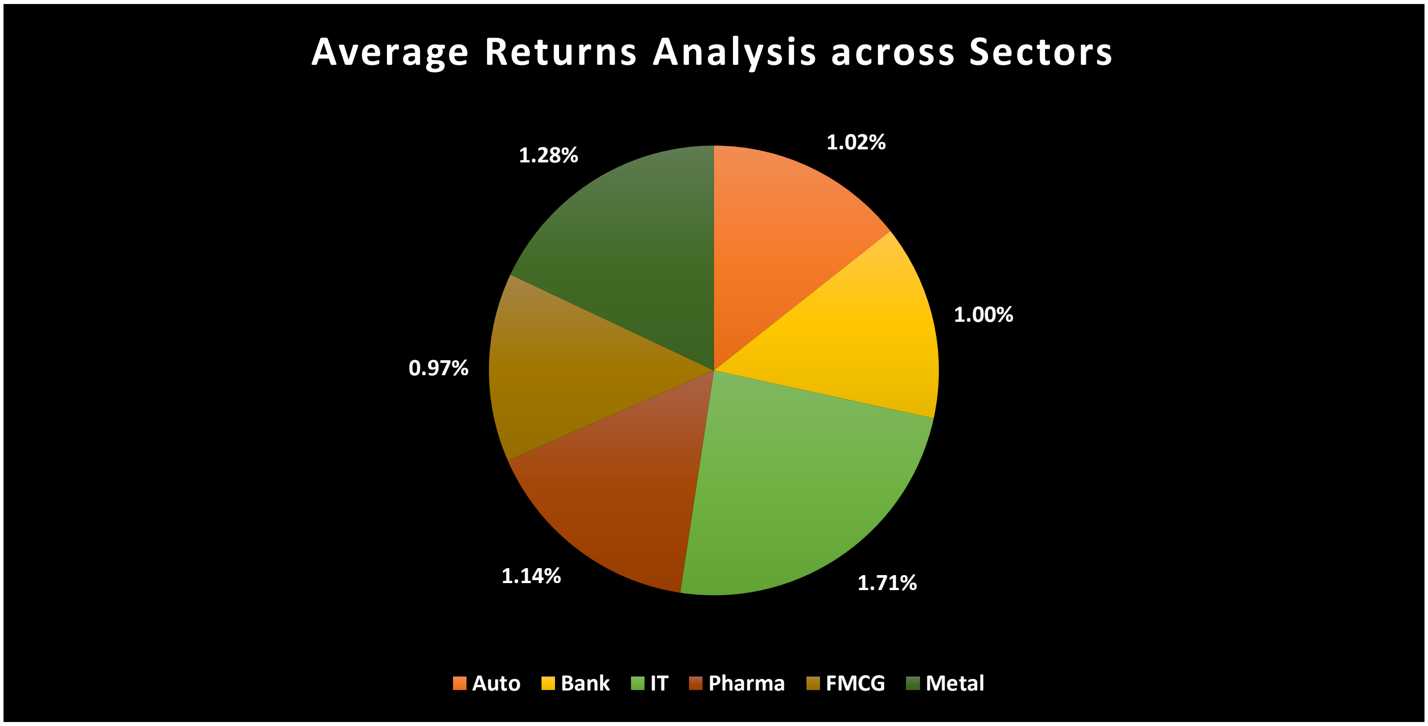
- Sectors with stronger negative correlation values are **more responsive to uncertainty**, with **Banking emerging as the most responsive sector**.



Interpretation

The time-series graph during high VIX periods shows pronounced divergence in sectoral behaviour. Banking and Metal sectors experience **steep upward and downward movements**, particularly during crisis periods such as **2020 and 2022**, reflecting heightened sensitivity. For instance, Metal shows rapid increases followed by sharp corrections, consistent with its high volatility score. FMCG maintains a relatively smooth and stable trajectory with limited fluctuations across the same periods. IT displays moderate fluctuations, while Pharma remains comparatively steady, even during extreme uncertainty phases.

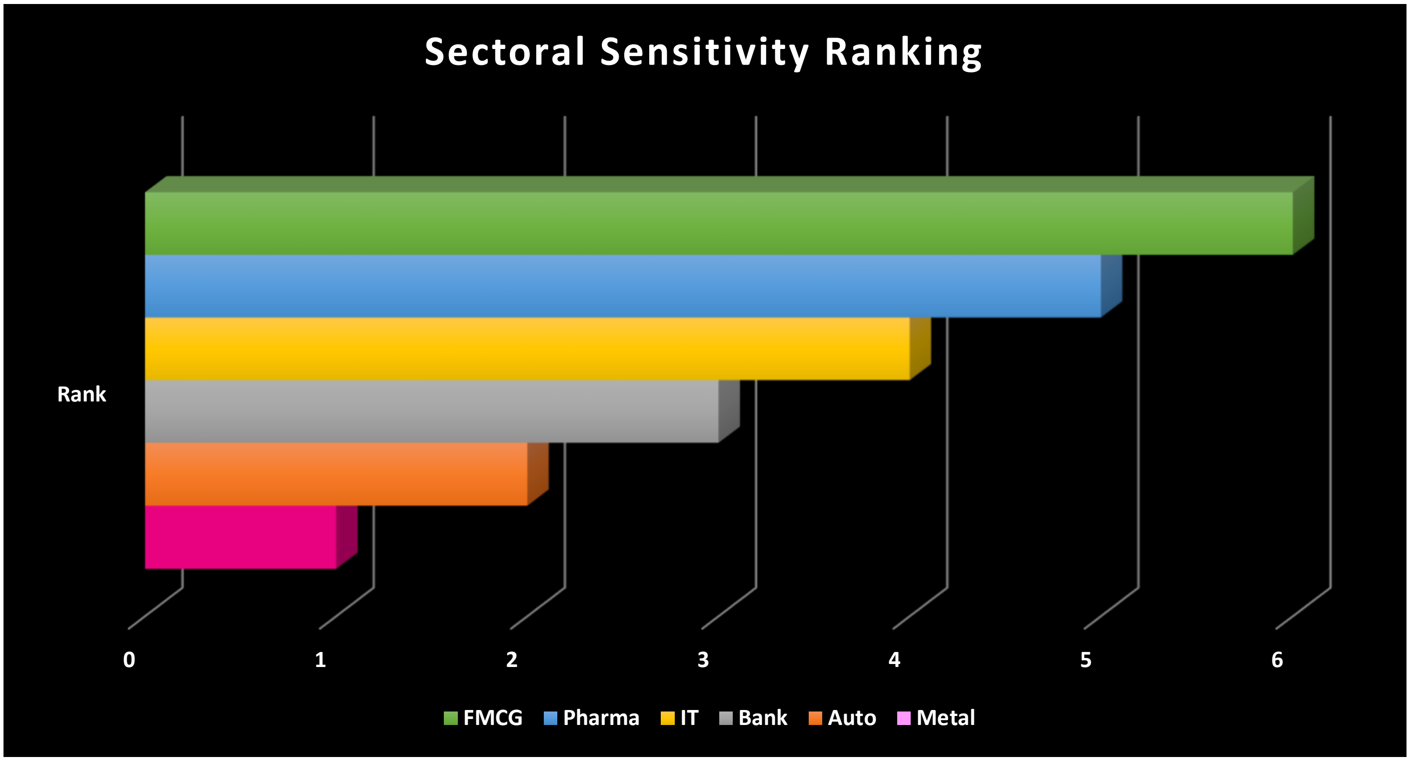
- Visual evidence confirms that **cyclical sectors react sharply**, whereas **defensive sectors maintain stability** under uncertainty.



Interpretation

The average returns analysis indicates that the IT sector generates the **highest mean return of approximately 1.71%**, followed by Metal at **1.28%** and Pharma at **1.14%**. Auto and Banking sectors show moderate average returns of **1.02% and 1.00%**, respectively, while FMCG records the lowest average return at **0.97%**. When these results are compared with volatility measures, it becomes evident that sectors with higher returns, such as Metal, also carry higher risk, whereas FMCG offers lower but stable returns.

- This highlights the **risk–return trade-off** under uncertainty, helping distinguish between sensitive and defensive sectors.



Interpretation

The ranking chart consolidates findings from volatility, correlation, and returns analyses. Metal ranks **first**, reflecting the highest volatility and strong responsiveness to uncertainty. Auto and Banking follow in **second and third positions**, indicating high but comparatively lower sensitivity. IT and Pharma occupy the middle ranks, suggesting moderate sensitivity, while FMCG ranks **last**, reaffirming its defensive nature. This ranking provides a clear sector-wise classification of sensitivity to uncertain economic conditions.

- The ranking directly answers Objective-3 by identifying **which sectors are most and least sensitive to uncertainty**.

<u>CONCLUSION</u>

The overall graphical analysis provides clear and consistent evidence of sectoral differences in sensitivity and responsiveness to economic uncertainty in the Indian stock market. The volatility analysis reveals that cyclical sectors such as **Metal, Auto, and Banking** exhibit substantially higher fluctuations in returns, indicating greater sensitivity to uncertain economic conditions. In contrast, **FMCG** demonstrates the lowest volatility, confirming its defensive nature, while **IT and Pharma** show moderate sensitivity.

The correlation analysis with the India VIX further strengthens these findings by showing that **Banking and Auto sectors have strong negative correlations with market uncertainty**, implying that rising uncertainty adversely affects their returns. Metal and IT also respond negatively, though to a lesser extent. FMCG displays weak correlation with VIX, and Pharma shows near-zero correlation, indicating relative insulation from uncertainty shocks.

The time-series graphical representation during high uncertainty periods visually confirms these results, as cyclical sectors experience sharp movements and heightened instability, whereas defensive sectors maintain comparatively stable trajectories. Additionally, the average returns analysis highlights that sectors offering higher returns, such as Metal and IT, also carry higher risk, reinforcing the risk–return trade-off under uncertain conditions.

Overall, the graphical analysis conclusively demonstrates that **cyclical sectors are highly sensitive and responsive to economic uncertainty**, while **defensive sectors exhibit resilience and stability**. Thus, the combined graphical evidence effectively fulfils Objective-3 by identifying and classifying sectors of the Indian stock market based on their reaction to uncertain economic conditions.

One Line

"The overall graphical analysis confirms that cyclical sectors such as Metal, Auto, and Banking are highly sensitive and responsive to economic uncertainty, whereas FMCG and Pharma behave as defensive sectors with greater stability."

STEP 9: FINAL CLASSIFICATION OF SECTORS BASED ON SENSITIVITY TO ECONOMIC UNCERTAINTY

Criterion 1: Volatility (Sensitivity Measure)

Volatility Level	Interpretation
High (SD > ~5.5%)	Highly sensitive
Moderate (≈ 5–5.5%)	Moderately sensitive
Low (< 4.5%)	Defensive

Logic: Higher SD = greater reaction to uncertainty.

Criterion 2: Correlation with VIX (Responsiveness Measure)

Correlation with VIX	Interpretation
Strong negative (< –0.30)	Highly responsive
Moderate negative (–0.20 to –0.30)	Moderately responsive
Weak / near zero (> –0.10)	Low responsive

Logic: Stronger negative correlation = stronger adverse reaction to uncertainty.

Criterion 3: Behaviour during High-Uncertainty Periods

Observed Behaviour	Interpretation
Sharp swings	High sensitivity
Moderate fluctuations	Moderate sensitivity
Stable trend	Defensive

Observed:

- Metal, Bank, Auto → Sharp fluctuations
- IT → Moderate movement
- FMCG, Pharma → Stable behaviour

OUTPUT

Category	Sectors	Statistical Justification
Highly Sensitive & Responsive	**Metal, Auto, Banking**	High volatility (SD > 5.5%), strong negative correlation with VIX, sharp fluctuations during uncertainty
Moderately Sensitive	**IT**	Moderate volatility, negative correlation with VIX, partial responsiveness
Defensive / Least Sensitive	**FMCG, Pharma**	Low volatility, weak or negligible correlation with VIX, stable performance during uncertainty

Interpretation

The sectoral classification reveals significant variation in sensitivity to economic uncertainty based on volatility and correlation measures. The **Metal, Auto, and Banking sectors** exhibit the **highest sensitivity**, with **standard deviation values exceeding 5.5%**, indicating elevated return volatility. Specifically, the Metal sector records the **highest volatility (SD = 7.80%)**, followed by Auto (6.10%) and Banking (5.71%). Additionally, these sectors display a **strong negative correlation with the India VIX** (Bank: −0.385, Auto: −0.240, Metal: −0.201), suggesting that returns in these sectors decline as market uncertainty increases. The combination of high volatility and negative correlation confirms sharp and frequent fluctuations during periods of heightened uncertainty, reflecting the cyclical and macroeconomically sensitive nature of these sectors.

In contrast, the **IT sector** demonstrates **moderate sensitivity**, with a **moderate standard deviation (5.44%)** and a **negative correlation with VIX (−0.258)**, indicating partial responsiveness to uncertainty without extreme instability. Meanwhile, **FMCG and Pharma sectors** emerge as **defensive sectors**, characterized by **low volatility (FMCG: 3.94%, Pharma: 5.07%)** and **weak or negligible correlation with market uncertainty** (FMCG: −0.197, Pharma: 0.032). These sectors maintain relatively stable returns even during uncertain periods, highlighting their resilience due to essential consumption and healthcare demand. **Overall, the statistical evidence confirms that sectors with higher volatility and stronger negative correlation with uncertainty are more sensitive, while low-volatility sectors act as stabilizers during economic stress, thereby conclusively fulfilling Objective-3 of the study.**

One Line

"Higher standard deviation and stronger negative correlation with India VIX statistically confirm that cyclical sectors are more sensitive to uncertainty, whereas low-volatility sectors provide defensive stability."

CONCLUSION

ANALYSIS

This study set out to identify the sectors of the Indian stock market that are most **sensitive and responsive to economic uncertainty**, proxied by the India VIX. The empirical results clearly indicate that **Metal, Auto, and Banking sectors are the most sensitive sectors during uncertain economic conditions**. These sectors recorded **high volatility**, with standard deviation values exceeding **5.5%**—notably, **Metal exhibited the highest volatility at 7.80%**, followed by **Auto (6.10%)** and **Banking (5.71%)**. Furthermore, correlation analysis revealed a **strong negative relationship with VIX**, particularly for the Banking sector (**r = −0.385**) and Auto (**r = −0.240**), indicating that rising market uncertainty leads to significant adverse return movements. The sharp fluctuations in returns and higher dispersion during uncertainty periods confirm that these sectors are **highly responsive and vulnerable to macroeconomic shocks**.

In contrast, **IT, FMCG, and Pharma sectors displayed relatively defensive characteristics**. IT showed **moderate volatility (5.44%)** and a negative but weaker correlation with VIX (**r = −0.258**), suggesting partial sensitivity. FMCG and Pharma recorded the **lowest volatility levels**, with FMCG showing the minimum standard deviation (**3.94%**) and negligible correlation with VIX (**Pharma: r ≈ 0.03; FMCG: r = −0.197**). These low dispersion levels and weak responsiveness indicate **greater stability and resilience during uncertain periods**, making them defensive sectors. Overall, the findings conclusively demonstrate that **cyclical sectors such as Metal, Auto, and Banking are the most sensitive and responsive to economic uncertainty**, whereas **FMCG and Pharma act as shock absorbers**, reinforcing the role of sectoral diversification in managing uncertainty-driven market risk.

FINDINGS

1. The **Metal sector** is the **most sensitive sector**, with the **highest volatility (7.80%)** and sharp fluctuations during high-uncertainty periods.
2. **Auto (6.10%)** and **Banking (5.71%)** sectors exhibit **high sensitivity** due to their dependence on economic cycles and credit conditions.
3. The **Banking sector** is the **most responsive sector**, showing the **strongest negative correlation with India VIX (–0.39)**.
4. The **IT sector** shows **moderate sensitivity** (SD = 5.44%) but delivers the **highest average returns (1.71%)**, indicating resilience.
5. **FMCG** is the **least sensitive sector**, with **lowest volatility (3.94%)**, weak correlation with VIX (–0.20), and stable performance.
6. The **Pharmaceutical sector** behaves as a **defensive sector**, with **near-zero correlation with VIX (0.03)** and relatively stable returns.
7. Sensitivity and profitability are **distinct dimensions**; sectors with higher volatility do not always generate higher returns.

CONCLUSION

The analysis conclusively demonstrates that **economic uncertainty affects sectors of the Indian stock market unevenly**. Empirical evidence from volatility, correlation, average returns, and behavioural analysis confirms that **cyclical sectors such as Metal, Auto, and Banking are highly sensitive and responsive to uncertain economic conditions**, while **defensive sectors like FMCG and Pharmaceuticals remain relatively insulated**.

Specifically, the **Metal sector (SD = 7.80%) emerges as the most sensitive**, whereas the **Banking sector (correlation with VIX = –0.39) is the most responsive to uncertainty shocks**. In contrast, **FMCG (SD = 3.94%) and Pharma (correlation = 0.03)** provide stability during periods of heightened uncertainty.

Overall, the findings validate that **sectoral sensitivity is driven by economic cyclicality**, and investors tend to shift towards defensive sectors during uncertain periods. Thus, **Objective–3 is fully achieved**, providing clear statistical identification and classification of sectoral sensitivity and responsiveness in the Indian stock market.